

Literature Status for Two p -Free-Space Reductions in arXiv:2011.12993

Run fa_banach_001

2026-07-01

Source

Fernando Albiac, Jose L. Ansorena, Marek Cuth, and Michal Doucha, *Lipschitz algebras and Lipschitz-free spaces over unbounded metric spaces*, arXiv:2011.12993.

The relevant source passages are Remark 2.8 and Remark 2.10. Remark 2.8 recalls the open problem whether the canonical linearization

$$L_j : \mathcal{F}_p(N) \longrightarrow \mathcal{F}_p(M)$$

of an inclusion $j : N \subset M$ is an isomorphism for $0 < p < 1$, and notes that the paper's bounded-space construction reduces the unbounded metric case to bounded metric spaces. Remark 2.10 recalls the open problem whether

$$E_{p,1,M} : \mathcal{F}_p(M) \longrightarrow \mathcal{F}(M)$$

is one-to-one for every metric space M and $0 < p < 1$, again reducing the issue to bounded metric spaces.

Supporting Literature

Fernando Albiac and Jose L. Ansorena, *Lipschitz extensions into p -Banach spaces, and canonical embeddings of Lipschitz-free p -spaces for $0 < p < 1$* , arXiv:2603.28193.

The abstract states that inclusions of p -metric spaces always induce genuine linear embeddings of Lipschitz-free p -spaces, and that the natural envelope map from $\mathcal{F}_p(M)$ to its q -Banach envelope $\mathcal{F}_q(M)$ is one-to-one for $0 < p < q \leq 1$.

Identification

The source's two reductions are answered by stronger later theorems:

- Theorem 3.13 of arXiv:2603.28193 proves that for every $0 < p \leq 1$ there is a constant $C(p)$ such that, for every pointed p -metric space M and every $0 \in N \subset M$, the canonical map $L[N, M; p] : \mathcal{F}_p(N) \rightarrow \mathcal{F}_p(M)$ is an isomorphic embedding.
- The final theorem of arXiv:2603.28193 proves that for every $0 < p \leq q \leq 1$ and every pointed q -metric space M , the canonical envelope map $J_{p,q}[M]$ is one-to-one. Taking $q = 1$ gives the injectivity of $E_{p,1,M} : \mathcal{F}_p(M) \rightarrow \mathcal{F}(M)$ asked about in the source.

Status

This is a literature-implied answer for arXiv:2011.12993: the later theorem solves, in stronger generality, the two older open problems whose unbounded-to- bounded reductions are recorded in the source paper. It is not a new proof packet. The first item overlaps the existing run packet `1811.01265_canonical_free_p_embedding_answered.2603.28193`; this packet is included to clear arXiv:2011.12993's own queue signal.